The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

$[P,T,x_s]$ If $\underline{t}$ = 4, the next three words are interpreted as pressure (Pa, lb_f/in^2), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

$[T,x_s,x_n]$ If $\underline{t}$ = 5, the next three words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

$[P,U_f,U_g,\alpha_g,x_n]$ If $\underline{t}$ = 6, the next five words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($\underline{t}$ = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then the vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy.

W2-W7(R) Quantities as described under Word 1 (W1). Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass liquid) follows the last required word for thermodynamic conditions.

8.3 Time-Dependent Volume Component, TMDPVOL

This component is indicated by TMDPVOL for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the volume in the time-dependent volume component is numbered as CCC010000.

8.3.1 Time-Dependent Volume Geometry, CCC0101-0109

This card (or cards) is required for a time-dependent volume component. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2, ft^2). When a time-dependent volume is used to model a pressure boundary condition (i.e., the time-dependent volume is connected to the system through a normal junction), it is generally recommended that the cross-sectional area of the time-dependent volume be large compared to the area of the normal junction.

W2(R) Length (m, ft). After initialization, the length is set to 0.

W3(R) Volume (m^3, ft^3). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the area times the length within a relative error of 0.000001. After initialization, the volume is set to 0.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. As with the other components, this Word 6 is compared to the length (Word 2) to determine if the horizontal or vertical flow regime map is used. This is not important for this component, since the correlations that depend on the flow regime maps are not needed for this component. The volume conditions are prescribed through input cards CCC0201-0299. After initialization, the elevation change is set to 0.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$.

If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made to ensure the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.

W9(I) Volume control flags. This word has the format tlpvbfe. For the time-dependent volume component, the volume control flag is restricted to the format 0000000. So enter 0 for this word.

8.3.2 Time-Dependent Volume Data Control Word, CCC0200

This card is required for a time-dependent volume.

W1(I) Control word for time-dependent data on CCC02NN cards. This word has the format εbt. It is not necessary to input leading zeros.

e-flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., ε = 1 specifies H_2O, ε = 2 specifies D_2O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-9 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words within the hydrodynamic system must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default ε = 0, then H_2O is assumed as the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit b specifies whether boron is present or not. The digit b = 0 specifies that the volume liquid does not contain boron; b = 1 specifies that a boron concentration in mass of boron per mass of liquid water (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit t specifies how the words of the time-dependent volume data in cards CCC0201-0299 are to be used to determine the initial thermodynamic state. Entering t equal to 0-3 specifies one component (steam/water). Entering t equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P,U_f,U_g,\alpha_g]$ If t = 0, the second, third, fourth, and fifth words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If t = 1, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as temperature (K, ^oF) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

[P,x_s] If t = 2, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

[P,T] If t = 3, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, ^oF) in equilibrium conditions. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E\text{-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\ E\text{-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If t = 4, the second, third, and fourth words of the time-dependent data on cards CCC0201-0299 are interpreted as pressure (Pa, $lb_f/in.^2$), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W5 if needed.

[T, x_s, x_n] If t = 5, the second, third, and fourth words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out. Enter boron in W5 if needed.

[$P, U_f, U_g, \alpha_g, x_n$] If t = 6, the second, third, fourth, fifth, and sixth words of the time-dependent data on cards CCC0201-0299 are interpreted as pressure (Pa, $lb_f/in.^2$), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present, and the input processing branches to that type of processing (t = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then the vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions.

W2(I) Table trip number. This word is optional. If missing or 0 and Word 3 is missing, no trip is used, and the time argument is the advancement time. If nonzero and Word 3 is missing, this number is the trip number, and the time argument is -1.0 if the trip is false, and the advancement time minus the trip time if the trip is true.

W3(A) Alphanumeric part of variable request code. This quantity is optional. If not present, time is the search argument. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. If the trip number is 0, the specified argument is used. If the trip number is nonzero, -1.0E+75 is used if the trip is false, and the specified argument is used if the trip is true. TIME can be selected, but note that the trip logic is different than if this word were omitted. The variable MFLOWJ should not be used as a search variable; a user-initialized control variable that uses MFLOWJ should be used instead.

W4(I) Numeric part of variable request code. This is assumed 0 if missing. For example, to use the pressure in volume 567010000 for the search argument, the user would enter P for W3 and 567010000 in W4.

8.3.3 Time-Dependent Volume Data, CCC0201-0299

These cards are required for time-dependent volume components. A set of data is made up of the search variable (e.g., time) followed by the required data indicated by control Word 1 in card CCC0200. The card numbers need not be consecutive, but the value of the search variable in a succeeding set must be equal to or greater than the value in the previous set. One or more sets of data, up to 5,000 sets, are allowed. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Linear interpolation is used if the search argument lies between the search variable entries. End-point values are used if the argument lies outside the table values. Only one set is needed if constant values are desired, and computer time is reduced when only one set is entered. Step changes can be accommodated by entering the two adjacent sets with the same search variable values or an extremely small difference between them. Given two identical argument values, the set selected will be the closest to the previous argument value. Sets may be entered one or more per card and may be split across cards. The total number of words must be a multiple of the set size.

Inputting time-dependent volume tables where the search variable is a thermodynamic variable from some other component can run into difficulties if the component numbering is such that the time-dependent volume is initialized before the component providing the needed search variable. A reliable fix for this is to make the search variable a control system output in the desired units, while the thermodynamic variable is the control system input in code internal (SI) units. The control system initial value can be set to the desired initial value of the search variable, and this will be used by the time-dependent table.

W1(R) Search variable (e.g., time).

W2-W7(R) Quantities as described under Word 1 in card 200. Depending on the control word, 2-5 quantities may be required. If entered, boron concentration (parts of boron per parts of liquid) follows the last required word for thermodynamic conditions.

As described above, sets may be entered one or more per card.

8.4 Single-Junction Component, SNGLJUN

A single-junction component is indicated by SNGLJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the single-junction component is numbered CCC000000.